

The two-measure functional Minkowski problem

Literature-already-answered note for arXiv:2105.09168

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Status. Completely answered by Falah–Rotem, arXiv:2502.16929, Theorem 1.5. The later paper gives necessary and sufficient conditions for a pair of measures to be the two surface-area measures of an integrable log-concave function, and proves uniqueness up to translation. This is precisely Problem 6.7 of the source paper, not merely a related one-measure or smooth special case.

The source question

For an upper-semicontinuous integrable log-concave function $g = e^{-\phi}$ on $\mathbb{R}^n$, the source associates two finite measures:

$$\mu_g = (\nabla\phi)_\#(g\,dx) \quad \text{on } \mathbb{R}^n, \quad \nu_g = (n_{\text{supp } g})_\#(g\,\mathcal{H}^{n-1}|_{\partial\text{supp } g}) \quad \text{on } \mathbb{S}^{n-1}.$$

After treating the essentially continuous case $\nu_g = 0$, Problem 6.7 on source PDF page 21 asks:

Fix a finite Borel measure μ on $\mathbb{R}^n$ and a finite Borel measure ν on $\mathbb{S}^{n-1}$. Under what conditions can one find $g \in \text{LC}_n$ with $\mu_g = \mu$ and $\nu_g = \nu$?

The surrounding text says that solving this problem is needed to extend the Riesz-type first-variation representation beyond the essentially continuous class.

The exact later theorem

Theorem 1.5 of arXiv:2502.16929 (supporting PDF page 5; source label `thm:main-Minkowski`) states the following. Let μ be a finite Borel measure on $\mathbb{R}^n$ and ν a finite Borel measure on $\mathbb{S}^{n-1}$. There is an integrable log-concave function f such that $(\mu_f, \nu_f) = (\mu, \nu)$ if and only if:

1. $\mu \neq 0$;
2. μ has finite first moment and the pair is centered, meaning

$$\int_{\mathbb{R}^n} x\,d\mu(x) + \int_{\mathbb{S}^{n-1}} \theta\,d\nu(\theta) = 0;$$

3. μ and ν are not supported on a common hyperplane $H \subseteq \mathbb{R}^n$.

Moreover, the realizing function is unique up to translation: if f and g have the same pair, then $g(x) = f(x + v)$ for some $v \in \mathbb{R}^n$.

The theorem’s “common hyperplane” condition uses the sphere as a subset of $\mathbb{R}^n$: there is no hyperplane $H \subseteq \mathbb{R}^n$ supporting both measures.

Why the identification is exact

Feature	Problem 6.7	Falah–Rotem Theorem 1.5
Interior measure	μ_g on $\mathbb{R}^n$	the same μ_f
Boundary measure	ν_g on $\mathbb{S}^{n-1}$	the same ν_f
Function class	integrable log-concave functions	the same class
Requested conclusion	characterize existence	iff conditions, plus uniqueness

The 2025 paper’s abstract also says explicitly that it “fully solve[s]” the functional Minkowski problem. Its theorem includes the earlier moment-measure case $\nu = 0$ but is not restricted to it. Thus the source problem is no longer open as stated.

Novelty status. This packet records an exact later-literature resolution. It does not claim a new theorem or an independent proof from this run.

Human review. Check Problem 6.7 on source PDF page 21 and Theorem 1.5 on supporting PDF page 5. The proof of the existence direction begins in Section 3 of the supporting paper; uniqueness is cited there as previously known in full generality.

References

- [1] L. Rotem, *A Riesz representation theorem for log-concave functions*, arXiv:2105.09168. See Problem 6.7 on PDF page 21.
- [2] T. Falah and L. Rotem, *On the functional Minkowski problem*, arXiv:2502.16929. See Theorem 1.5 on PDF page 5.